1	(a)		region of space area / volume	B1	
			where a mass experiences a force	B1	[2]
	(b)	(i)	force proportional to product of two masses	M1	
			force inversely proportional to the square of their separation	M1	
			either reference to point masses *or* separation >> 'size' of masses	A1	[3]
		(ii)	field strength $= GM / x^2$ **or** field strength $\propto 1 / x^2$	C1	
			ratio $= (7.78 \times 10^8)^2 / (1.5 \times 10^8)^2$	C1	
			$= 27$	A1	[3]
	(c)	(i)	*either* centripetal force $= mR\omega^2$ and $\omega = 2\pi / T$		
			or centripetal force $= mv^2 / R$ and $v = 2\pi R / T$	B1	
			gravitational force provides the centripetal force	B1	
			either $GMm / R^2 = mR\omega^2$ *or* $GMm / R^2 = mv^2 / R$	M1	
			$M = 4\pi^2 R^3 / GT^2$	A0	[3]
			(allow working to be given in terms of acceleration)		
		(ii)	$M = \{4\pi^2 \times (1.5 \times 10^{11})^3\} / \{6.67 \times 10^{-11} \times (3.16 \times 10^7)^2\}$	C1	
			$= 2.0 \times 10^{30}$ kg	A1	[2]